



Queensland University of Technology
Faculty of Science

Project Requirements

Team Champions – CAB302

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1. Introduction:

The project Challenge Hub is a gamified self-development application that turns personal growth into a personality-driven adventure. Users complete a short MBTI-style quiz that assigns them one of several archetypes, with each tight to the 12 Labors of Heracles. After that, the system shapes a personalized questline composes of daily challenges and reflections geared toward a goal that users choose for themselves. A daily quest cap keeps engagement sustainable rather than compulsive. Streaks, badges, friend feed, and radar chart of progress keep users motivated to return. The AI personalization layer make the experience increasing tailored over time.

2. Technical Requirements:

Front-end: JavaFX

Back-end: Java, JDBC

Database: SQLite

3. Scope:

The Challenge Hub software focuses on these scopes:

Feature 1	Onboarding & Personality Assessment
Feature 2	Core Progression Loop
Feature 3	Rewards & Social

4. Functional Requirements:

4.1. Onboarding & Personality Assessment:

- The system presents a 20-question onboarding quiz during user signup to assess MBTI-style traits
- Quiz shall assess only the Extraversion/Introversion (E/I) dimension
- The system maps quiz results to one of 12 user archetypes, each linked to one of 12 Labours of Heracles
- The system shall let user express a personal goal in relation to their results and use this to steer content

4.2. Core Progression Loop:

- The system generates a main questline tailored to user's onboarding result. Each quest contains narrative, a challenge, and a reflection prompt
- The system shall issue global daily quests shared across users of the same MBTI type
- The system break quests down into "Tasks", which are smally daily actions that feed into larger quest

- The system tracks and updates streak based on daily quest and task completion

4.3. Rewards & Social:

- The system awards badges for every completed quest
- The system displays a friend progress feed showing other user's activities
- The system shall provide "AI personalisation" that adapts contents/recommendations to individual users

4.4. User Management:

- Users must be able to register and log into the account
- Users must be able to view the profiles, charts, and badges
- Users must be able to do the MBTI-type quiz after they have registered the account
- Users must be able to do daily challenges and quests after they log in

5. Non-functional Requirements:

5.1. Architecture:

- Front-end must be built in JavaFX, with low-fidelities and medium-fidelities UI during design
- Backend must be implemented using Java, with JDBC for data access
- Data should be persisted in SQLite
- Software includes clear Test Driven Development, and CI/CD workflows for deployment

5.2. Usability:

- The onboarding quiz should be short enough to minimize drop-off during registration
- Reflection prompts and narrative content should be presented in a way that is approachable rather than clinical

5.3. Scalability:

- Data model and quiz logic should be designed to extend from E/I
- Content structure, which includes Quests, Dailies, Tasks, should be modular enough to support additional archetypes, labours, and story content

5.4. Performance:

- The radar chart and progress dashboard should update responsively as new quiz is recorded

5.5. Reliability:

- The daily engagement shall consistently and correctly limit users to one

5.6. Privacy:

- Personality data is sensitive so the system must define access, storage, and retention rules for it
- User's password must be hashed before storing in the database